

```
In [1]: !pip install ydata-profiling

import pandas as pd

from ydata_profiling import ProfileReport

# Now you can use the profiling tool
# Example usage:
# df = pd.DataFrame({'col1': [1, 2, 3], 'col2': [4, 5, 6]})
# profile = ProfileReport(df, title="Pandas Profiling Report")
# profile.to_file("output.html")
```

Collecting ydata-profiling

Using cached ydata_profiling-4.18.1-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: scipy<1.17,>=1.8 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (1.15.3)

Requirement already satisfied: pandas!=1.4.0,<3.0,>1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.2.3)

Requirement already satisfied: matplotlib<=3.10,>=3.5 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (3.10.0)

Requirement already satisfied: pydantic<3,>=2 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.10.3)

Collecting PyYAML<6.1,>=6.0.3 (from ydata-profiling)

Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl.metadata (2.4 kB)

Requirement already satisfied: jinja2<3.2,>=3.1.6 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (3.1.6)

Collecting visions<0.8.2,>=0.7.5 (from visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)

Using cached visions-0.8.1-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: numpy<2.4,>=1.22 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.1.3)

Collecting minify-html>=0.15.0 (from ydata-profiling)

Using cached minify_html-0.18.1-cp313-cp313-win_amd64.whl.metadata (18 kB)

Collecting filetype>=1.0.0 (from ydata-profiling)

Using cached filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)

Collecting phik<0.13,>=0.12.5 (from ydata-profiling)

Using cached phik-0.12.5-cp313-cp313-win_amd64.whl.metadata (5.6 kB)

Requirement already satisfied: requests<3,>=2.32.0 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (2.32.3)

Requirement already satisfied: tqdm<5,>=4.66.3 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (4.67.1)

Requirement already satisfied: seaborn<0.14,>=0.10.1 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.13.2)

Collecting multimethod<2,>=1.4 (from ydata-profiling)

Using cached multimethod-1.12-py3-none-any.whl.metadata (9.6 kB)

Requirement already satisfied: statsmodels<1,>=0.13.2 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.14.4)

Collecting typeguard<5,>=4 (from ydata-profiling)

Using cached typeguard-4.4.4-py3-none-any.whl.metadata (3.3 kB)

Collecting imagehash==4.3.2 (from ydata-profiling)

Using cached ImageHash-4.3.2-py2.py3-none-any.whl.metadata (8.4 kB)

Collecting wordcloud>=1.9.4 (from ydata-profiling)

Using cached wordcloud-1.9.6-cp313-cp313-win_amd64.whl.metadata (3.5 kB)

Collecting dacite<2,>=1.9 (from ydata-profiling)

Using cached dacite-1.9.2-py3-none-any.whl.metadata (17 kB)

Requirement already satisfied: numba<0.63,>=0.60 in c:\users\lenovo\anaconda3\lib\site-packages (from ydata-profiling) (0.61.0)

Requirement already satisfied: PyWavelets in c:\users\lenovo\anaconda3\lib\site-packages (from imagehash==4.3.2->ydata-profiling) (1.8.0)

Requirement already satisfied: pillow in c:\users\lenovo\anaconda3\lib\site-packages (from imagehash==4.3.2->ydata-profiling)

(11.1.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\lenovo\anaconda3\lib\site-packages (from jinja2<3.2,>=3.1.6->ydata-profiling) (3.0.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.3.1)

Requirement already satisfied: cycler>=0.10 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (4.55.3)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (1.4.8)

Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (24.2)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (3.2.0)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\lenovo\anaconda3\lib\site-packages (from matplotlib<=3.10,>=3.5->ydata-profiling) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.45,>=0.44.0dev0 in c:\users\lenovo\anaconda3\lib\site-packages (from numba<0.63,>=0.60->ydata-profiling) (0.44.0)

Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2024.1)

Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\site-packages (from pandas!=1.4.0,<3.0,>1.5->ydata-profiling) (2025.2)

Requirement already satisfied: joblib>=0.14.1 in c:\users\lenovo\anaconda3\lib\site-packages (from phik<0.13,>=0.12.5->ydata-profiling) (1.4.2)

Requirement already satisfied: annotated-types>=0.6.0 in c:\users\lenovo\anaconda3\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (0.6.0)

Requirement already satisfied: pydantic-core==2.27.1 in c:\users\lenovo\anaconda3\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (2.27.1)

Requirement already satisfied: typing-extensions>=4.12.2 in c:\users\lenovo\anaconda3\lib\site-packages (from pydantic<3,>=2->ydata-profiling) (4.12.2)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.32.0->ydata-profiling) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.32.0->ydata-profiling) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.32.0->ydata-profiling) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.32.0->ydata-profiling) (2025.8.3)

Requirement already satisfied: patsy>=0.5.6 in c:\users\lenovo\anaconda3\lib\site-packages (from statsmodels<1,>=0.13.2->ydata-profiling) (1.0.1)

```

Requirement already satisfied: colorama in c:\users\lenovo\anaconda3\lib\site-packages (from tqdm<5,>=4.66.3->ydata-profiling)
(0.4.6)
Collecting typing-extensions>=4.12.2 (from pydantic<3,>=2->ydata-profiling)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: attrs>=19.3.0 in c:\users\lenovo\anaconda3\lib\site-packages (from visions<0.8.2,>=0.7.5->visions
s[type_image_path]<0.8.2,>=0.7.5->ydata-profiling) (24.3.0)
Requirement already satisfied: networkx>=2.4 in c:\users\lenovo\anaconda3\lib\site-packages (from visions<0.8.2,>=0.7.5->visions
s[type_image_path]<0.8.2,>=0.7.5->ydata-profiling) (3.4.2)
Collecting puremagic (from visions<0.8.2,>=0.7.5->visions[type_image_path]<0.8.2,>=0.7.5->ydata-profiling)
  Using cached puremagic-1.30-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib<=
3.10,>=3.5->ydata-profiling) (1.17.0)
Using cached ydata_profiling-4.18.1-py2.py3-none-any.whl (400 kB)
Using cached ImageHash-4.3.2-py2.py3-none-any.whl (296 kB)
Using cached dacite-1.9.2-py3-none-any.whl (16 kB)
Using cached multimethod-1.12-py3-none-any.whl (10 kB)
Using cached phik-0.12.5-cp313-cp313-win_amd64.whl (678 kB)
Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl (154 kB)
Using cached typeguard-4.4.4-py3-none-any.whl (34 kB)
Using cached visions-0.8.1-py3-none-any.whl (105 kB)
Using cached filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Downloading minify_html-0.18.1-cp313-cp313-win_amd64.whl (3.1 MB)
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----- 0.0/3.1 MB ? eta -:-:--
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----- 1.3/3.1 MB 1.2 MB/s eta 0:00:02
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----- 2.6/3.1 MB 948.1 kB/s eta 0:00:01
----- 2.6/3.1 MB 948.1 kB/s eta 0:00:01
----- 2.9/3.1 MB 914.7 kB/s eta 0:00:01
----- 2.9/3.1 MB 914.7 kB/s eta 0:00:01
----- 3.1/3.1 MB 899.0 kB/s eta 0:00:00
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)

```


[illegible]

```

----- 4/13 [PyYAML]
----- 4/13 [PyYAML]
----- 4/13 [PyYAML]
----- 6/13 [dacite]
----- 9/13 [wordcloud]
----- 10/13 [visions]
----- 10/13 [visions]
----- 10/13 [visions]
----- 10/13 [visions]
----- 11/13 [phik]
----- 12/13 [ydata-profiling]
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----- 12/13 [ydata-profiling]
----- 12/13 [ydata-profiling]
----- 12/13 [ydata-profiling]
----- 13/13 [ydata-profiling]

```

Successfully installed PyYAML-6.0.3 dacite-1.9.2 filetype-1.2.0 imagehash-4.3.2 minify-html-0.18.1 multimethod-1.12 phik-0.12.5 puremagic-1.30 typeguard-4.4.4 typing-extensions-4.15.0 visions-0.8.1 wordcloud-1.9.6 ydata-profiling-4.18.1

WARNING: Failed to remove contents in a temporary directory 'C:\Users\LENOVO\anaconda3\Lib\site-packages\~aml'.
You can safely remove it manually.

```

In [2]: %%time
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
import pandas_profiling as prf

```

CPU times: total: 31.2 ms

Wall time: 5.69 ms

<timed exec>:4: DeprecationWarning: `import pandas\_profiling` is going to be deprecated by April 1st. Please use `import ydata\_profiling` instead.

```

In [6]: netflix = pd.read_csv("D:\\manish\\Python\\ineuron\\Anand Jha Drive for INEURON FSDA\\PYTHON\\Python\\EDA\\netflix_data.csv")
netflix.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 13 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Unnamed: 0      8807 non-null   int64
 1   show_id         8807 non-null   object
 2   type            8807 non-null   object
 3   title           8807 non-null   object
 4   director        6173 non-null   object
 5   cast            7982 non-null   object
 6   country         7976 non-null   object
 7   date_added      8797 non-null   object
 8   release_year    8807 non-null   int64
 9   rating          8803 non-null   object
10   duration        8804 non-null   object
11   listed_in       8807 non-null   object
12   description     8807 non-null   object
dtypes: int64(2), object(11)
memory usage: 894.6+ KB

```

About Data

This dataset contains data collected from Netflix of different TV shows and movies from the year 2008 to 2021.

- type: Gives information about 2 different unique values one is TV Show and another is Movie
- title: Gives information about the title of Movie or TV Show
- director: Gives information about the director who directed the Movie or TV Show
- cast: Gives information about the cast who plays role in Movie or TV Show
- release_year: Gives information about the year when Movie or TV Show was released
- rating: Gives information about the Movie or TV Show are in which category (eg like the movies are only for students, or adults, etc)
- duration: Gives information about the duration of Movie or TV Show
- listed_in: Gives information about the genre of Movie or TV Show
- description: Gives information about the description of Movie or TV Show


```
In [7]: m=netflix.isnull().sum()
miss= (netflix.isnull().sum()/len(netflix))*100
miss_data=pd.concat([m,miss],axis=1,keys=['Total','%'])
print(miss_data)
```

	Total	%
Unnamed: 0	0	0.000000
show_id	0	0.000000
type	0	0.000000
title	0	0.000000
director	2634	29.908028
cast	825	9.367549
country	831	9.435676
date_added	10	0.113546
release_year	0	0.000000
rating	4	0.045418
duration	3	0.034064
listed_in	0	0.000000
description	0	0.000000

Dropping the cast and director features because we are not going to use those features right now

```
In [8]: netflix = netflix.dropna( how='any',subset=['cast', 'director'])
```

```
In [9]: netflix.fillna({'country': 'missing', 'rating': 'missing'}, inplace=True)
netflix.isnull().sum()
```

```
Out[9]: Unnamed: 0      0
        show_id      0
        type         0
        title        0
        director     0
        cast         0
        country      0
        date_added   0
        release_year 0
        rating       0
        duration     3
        listed_in    0
        description  0
        dtype: int64
```

```
In [10]: netflix['date_added'] = pd.to_datetime(netflix['date_added'])
        netflix.head()
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_14916\3787484694.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
netflix['date_added'] = pd.to_datetime(netflix['date_added'])
```

Out[10]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	des
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To h c
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	TI ch pri
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Children & Family Movies	Ec div e
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On G A r
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	British TV Shows, Reality TV	A ba



```
In [11]: ## Finding how many unique values are there in the dataset  
netflix.nunique()
```

```
Out[11]: Unnamed: 0      5700  
show_id      5700  
type         2  
title        5698  
director     4152  
cast         5512  
country      604  
date_added   1478  
release_year  72  
rating       18  
duration     205  
listed_in    346  
description   5677  
dtype: int64
```

```
In [12]: netflix.describe(include='all')
```

Out[12]:

	Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	d
count	5700.000000	5700	5700	5700	5700	5700	5700	5700	5700.000000	5700	5697	5700	
unique	NaN	5700	2	5698	4152	5512	604	NaN	NaN	18	205	346	
top	NaN	s8807	Movie	22-Jul	Raúl Campos, Jan Suter	Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jig...	United States	NaN	NaN	TV-MA	94 min	Dramas, International Movies	
freq	NaN	1	5522	2	18	13	1849	NaN	NaN	1939	140	361	
mean	4551.407018	NaN	NaN	NaN	NaN	NaN	NaN	2019-05-28 02:41:56.210526208	2012.979474	NaN	NaN	NaN	
min	2.000000	NaN	NaN	NaN	NaN	NaN	NaN	2008-01-01 00:00:00	1942.000000	NaN	NaN	NaN	
25%	2276.750000	NaN	NaN	NaN	NaN	NaN	NaN	2018-05-04 00:00:00	2012.000000	NaN	NaN	NaN	
50%	4697.500000	NaN	NaN	NaN	NaN	NaN	NaN	2019-07-21 12:00:00	2016.000000	NaN	NaN	NaN	
75%	6828.250000	NaN	NaN	NaN	NaN	NaN	NaN	2020-08-13 00:00:00	2018.000000	NaN	NaN	NaN	
max	8806.000000	NaN	NaN	NaN	NaN	NaN	NaN	2021-09-24 00:00:00	2021.000000	NaN	NaN	NaN	
std	2590.214753	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.562133	NaN	NaN	NaN	

EDA

- What different types of show or movie are uploaded on Netflix?
- Correlation between the features
- Most watched shows on the Netflix
- Distribution of Ratings
- Which has the highest rating Tv show or Movies
- Finding the best Month for releasing content
- Highest watched genres on Netflix
- Released movie over the years

```
In [13]: ##method-1
netflix.groupby('type')['title'].count().sort_values(ascending = False)
```

```
Out[13]: type
Movie      5522
TV Show     178
Name: title, dtype: int64
```

```
In [14]: ##method-2
netflix.type.value_counts().to_frame('values_count')
```

```
Out[14]:
```

	values_count
type	
Movie	5522
TV Show	178

```
In [16]: numeric_columns = netflix.select_dtypes(include=['float64', 'int64']).columns

correlation_matrix = netflix[numeric_columns].corr()
```

```
correlation_matrix
```

```
Out[16]:
```

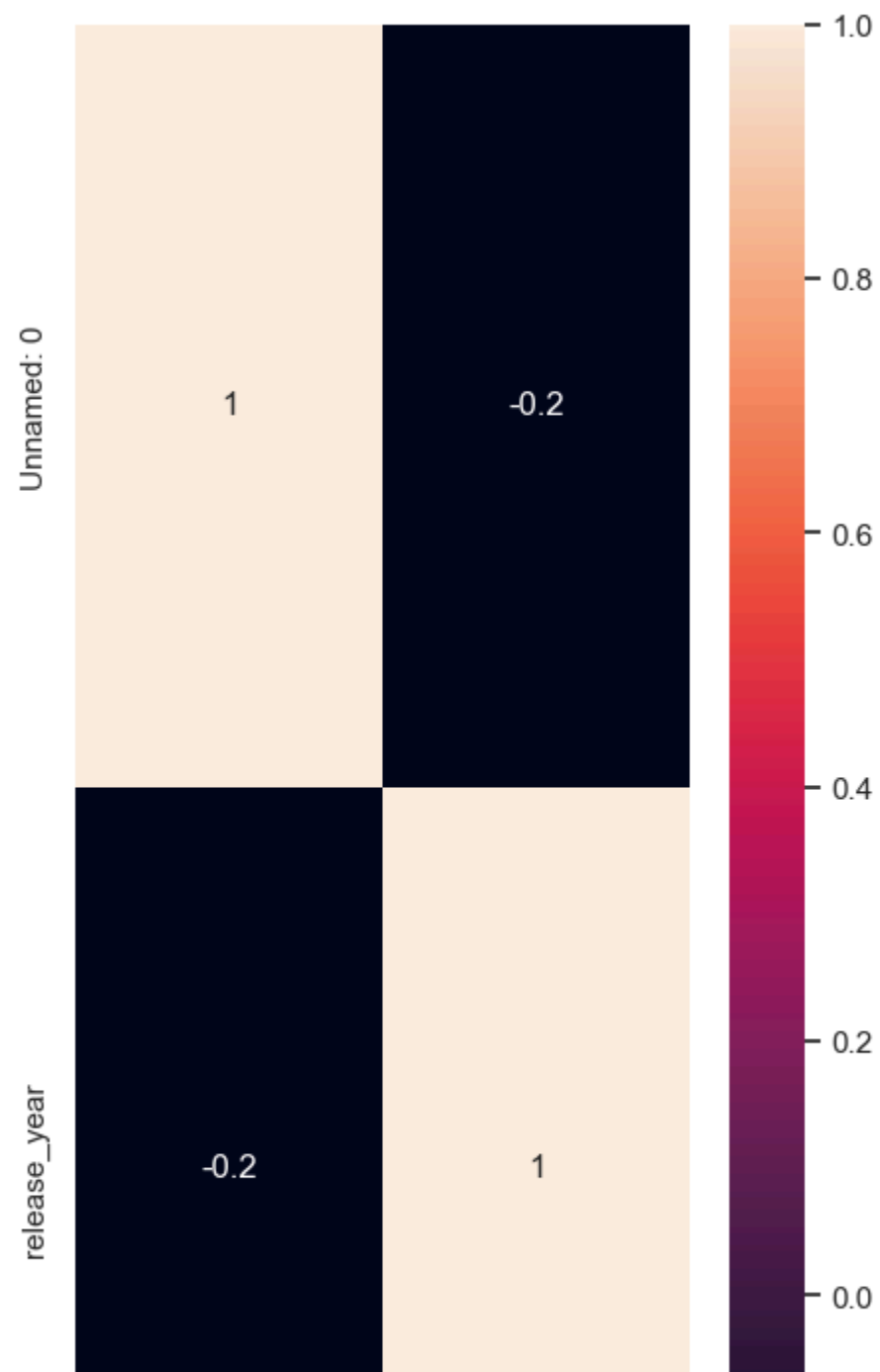
	Unnamed: 0	release_year
--	------------	--------------

Unnamed: 0	1.000000	-0.199716
------------	----------	-----------

release_year	-0.199716	1.000000
--------------	-----------	----------

```
In [18]: sns.set()  
plt.subplots(figsize=(5,10))  
sns.heatmap(correlation_matrix,annot=True)
```

```
Out[18]: <Axes: >
```





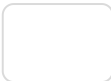
```
In [19]: netflix_profile = prf.ProfileReport(netflix)
netflix_profile
```

```
Summarize dataset: 0%|          | 0/5 [00:00<?, ?it/s]
```

Iteration	Time (s)	Throughput (it/s)
0/13	[00:00<?, ?it/s]	
1/13	[00:00<00:01, 9.03it/s]	9.03
2/13	[00:00<00:01, 8.13it/s]	8.13
3/13	[00:00<00:01, 8.87it/s]	8.87
5/13	[00:00<00:00, 12.94it/s]	12.94
7/13	[00:00<00:00, 6.67it/s]	6.67
13/13	[00:01<00:00, 9.58it/s]	9.58

```
Generate report structure: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
Render HTML:  0%|          | 0/1 [00:00<?, ?it/s]
```



Overview

Overview

Alerts 3

Reproduction

Dataset statistics

Number of variables	13
Number of observations	5700
Missing cells	3
Missing cells (%)	< 0.1%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	623.4 KiB
Average record size in memory	112.0 B

Variable types

Numeric	2
Text	8
Categorical	2
DateTime	1

Variables



Select Columns

Unnamed: 0

Out[19]:

```
In [22]: netflix_profile.to_file(output_file="Movies_before_preprocessing.html")
```

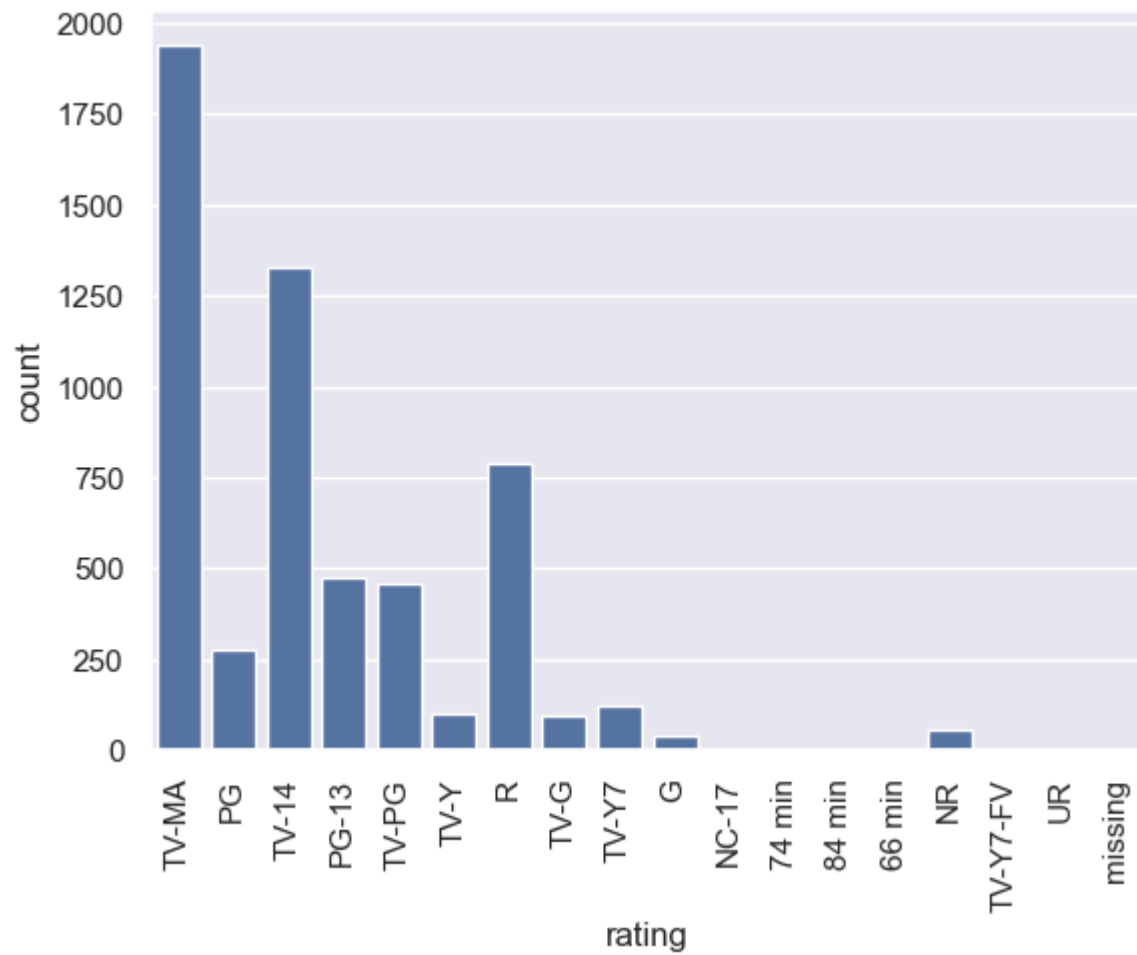
Export report to file: 0%| | 0/1 [00:00<?, ?it/s]

```
In [23]: netflix.groupby(["type", "release_year"])["title"].agg(pd.Series.mode)
```

```
Out[23]: type    release_year
Movie    1942                The Battle of Midway
         1944                Tunisian Victory
         1945            Know Your Enemy - Japan
         1946            Let There Be Light
         1947            Thunderbolt
         ...
TV Show  2017    [Abnormal Summit, Afronta! Facing It!, Ari Sha...
         2018    [A.I.C.O., Age of Rebellion, Bitter Daisies, B...
         2019    [7 (Seven), A Perfect Day For Arsenide, Abby H...
         2020    [A Queen Is Born, Almost Happy, Anitta: Made I...
         2021    [Bangkok Breaking, Clickbait, Dealer, Feel Goo...
Name: title, Length: 88, dtype: object
```

```
In [24]: sns.countplot(x=netflix['rating'],orient = 'v')
plt.xticks(rotation = 90)
```

```
Out[24]: ([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17],
[Text(0, 0, 'TV-MA'),
Text(1, 0, 'PG'),
Text(2, 0, 'TV-14'),
Text(3, 0, 'PG-13'),
Text(4, 0, 'TV-PG'),
Text(5, 0, 'TV-Y'),
Text(6, 0, 'R'),
Text(7, 0, 'TV-G'),
Text(8, 0, 'TV-Y7'),
Text(9, 0, 'G'),
Text(10, 0, 'NC-17'),
Text(11, 0, '74 min'),
Text(12, 0, '84 min'),
Text(13, 0, '66 min'),
Text(14, 0, 'NR'),
Text(15, 0, 'TV-Y7-FV'),
Text(16, 0, 'UR'),
Text(17, 0, 'missing')])
```



```
In [25]: netflix.groupby("type")["rating"].agg(pd.Series.mode)
```

```
Out[25]: type
Movie    TV-MA
TV Show  TV-MA
Name: rating, dtype: object
```

```
In [26]: netflix.head()
```

Out[26]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	des
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To h c
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	TI ch pri
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Children & Family Movies	Ec div e
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On G A r
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	British TV Shows, Reality TV	A ba

In [30]: new_genre = netflix['listed_in'].str.split(",",n=2)

```
In [31]: new_genre
```

```
Out[31]: 2      [Crime TV Shows, International TV Shows, TV ...
5          [TV Dramas, TV Horror, TV Mysteries]
6          [Children & Family Movies]
7      [Dramas, Independent Movies, International M...
8          [British TV Shows, Reality TV]
...
8801      [Dramas, International Movies, Thrillers]
8802      [Cult Movies, Dramas, Thrillers]
8804      [Comedies, Horror Movies]
8805      [Children & Family Movies, Comedies]
8806      [Dramas, International Movies, Music & Music...
Name: listed_in, Length: 5700, dtype: object
```

Adding new features

```
In [32]: netflix['Genre 1'] = new_genre.str.get(0)
netflix['Genre 2'] = new_genre.str.get(1)
netflix['Genre 3'] = new_genre.str.get(2)
```

```
In [33]: netflix.head()
```

Out[33]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	des
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To h c
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	TI ch pri
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Children & Family Movies	Ec div e
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On G A r
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	British TV Shows, Reality TV	A ba

In [34]: `netflix['Genre 1'].describe(include='all')`


```
Out[34]: count      5700  
         unique      33  
         top      Dramas  
         freq      1586  
         Name: Genre 1, dtype: object
```

```
In [35]: netflix['Genre 2'].describe(include='all')
```

```
Out[35]: count      4485  
         unique      35  
         top      International Movies  
         freq      1545  
         Name: Genre 2, dtype: object
```

```
In [36]: netflix['Genre 3'].describe(include='all')
```

```
Out[36]: count      2434  
         unique      29  
         top      International Movies  
         freq      881  
         Name: Genre 3, dtype: object
```

```
In [37]: netflix.drop('listed_in',axis = 1,inplace = True)
```

```
In [38]: netflix.head()
```

Out[38]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description	Genre
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	To protect his family from a powerful drug lor...	Crime, Drama
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	The arrival of a charismatic young priest brin...	Drama
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Equestria's divided. But a bright-eyed hero be...	Children's, Fantasy, Music
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	On a photo shoot in Ghana, an American model s...	Drama
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	A talented batch of amateur bakers face off in...	Baking, Reality, Show

In [39]: `netflix.drop_duplicates(subset = None,keep='first',inplace=False)`

Out[39]:

[illegible]

	Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description
8801	8801	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...	United Arab Emirates, Jordan	2016-03-09	2015	TV-MA	96 min	Recovering alcoholic Talal wakes up inside a s...
8802	8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	A political cartoonist, a crime reporter and a...
8804	8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Looking to survive in a world taken over by zo...
8805	8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Dragged from civilian life, a former superhero...
8806	8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	A scrappy but poor boy worms his way into a ty...

5700 rows × 15 columns

Q1) Movies made on year basis?

```
In [40]: netflix.groupby(['Genre 1'])['release_year'].count().sort_values(ascending = False)
```

```
Out[40]: Genre 1
Dramas                1586
Comedies              1184
Action & Adventure    848
Children & Family Movies 548
Documentaries         397
Stand-Up Comedy       309
Horror Movies         274
International Movies   122
Classic Movies         73
Thrillers             65
International TV Shows 59
Crime TV Shows        37
Movies               30
British TV Shows      21
Independent Movies     20
Anime Features         19
Music & Musicals       18
Sci-Fi & Fantasy       13
Cult Movies            12
Anime Series          11
Kids' TV              11
Docuseries            10
TV Shows              9
Stand-Up Comedy & Talk Shows 8
TV Comedies           3
Romantic Movies       3
Classic & Cult TV      2
TV Dramas             2
TV Action & Adventure  2
Romantic TV Shows     1
Reality TV            1
LGBTQ Movies          1
TV Horror             1
Name: release_year, dtype: int64
```

```
In [41]: netflix.groupby(['Genre 2'])['release_year'].count().sort_values(ascending = False)
```

```
Out[41]: Genre 2
          International Movies      1545
          Dramas                    744
          Independent Movies        572
          Comedies                   440
          Thrillers                  232
          Romantic Movies           170
          Music & Musicals           119
          Sci-Fi & Fantasy           109
          Horror Movies              71
          Sports Movies              68
          Anime Features             49
          Faith & Spirituality       49
          Cult Movies                44
          International TV Shows     41
          Children & Family Movies   31
          Classic Movies             30
          LGBTQ Movies              30
          TV Dramas                  23
          Romantic TV Shows          22
          Documentaries              21
          TV Comedies                20
          Stand-Up Comedy            9
          Korean TV Shows            9
          Crime TV Shows             8
          TV Action & Adventure       7
          Docuseries                 7
          TV Horror                   3
          Classic & Cult TV           2
          Kids' TV                    2
          Reality TV                  2
          TV Sci-Fi & Fantasy         2
          Science & Nature TV        1
          Spanish-Language TV Shows  1
          Stand-Up Comedy & Talk Shows 1
          TV Mysteries               1
          Name: release_year, dtype: int64
```

```
In [42]: netflix.groupby(['Genre 3'])['release_year'].count().sort_values(ascending = False)
```

```
Out[42]: Genre 3
          International Movies      881
          Romantic Movies          431
          Thrillers                 278
          Music & Musicals          191
          Independent Movies        159
          Sci-Fi & Fantasy          115
          Sports Movies             95
          Dramas                   68
          LGBTQ Movies              52
          TV Dramas                 36
          TV Comedies               16
          Cult Movies               14
          Comedies                  12
          Faith & Spirituality       11
          Spanish-Language TV Shows 10
          TV Mysteries              10
          TV Action & Adventure       7
          International TV Shows      7
          Horror Movies              7
          Classic Movies             6
          TV Horror                  5
          Romantic TV Shows          5
          Children & Family Movies    4
          Stand-Up Comedy & Talk Shows 3
          TV Sci-Fi & Fantasy          3
          TV Thrillers               3
          Teen TV Shows              3
          Korean TV Shows            1
          Kids' TV                   1
          Name: release_year, dtype: int64
```

```
In [43]: ##no of movies & tv shows released by year arranged in descedning order
netflix.groupby(['release_year'])['release_year'].count().sort_values(ascending=False)
```

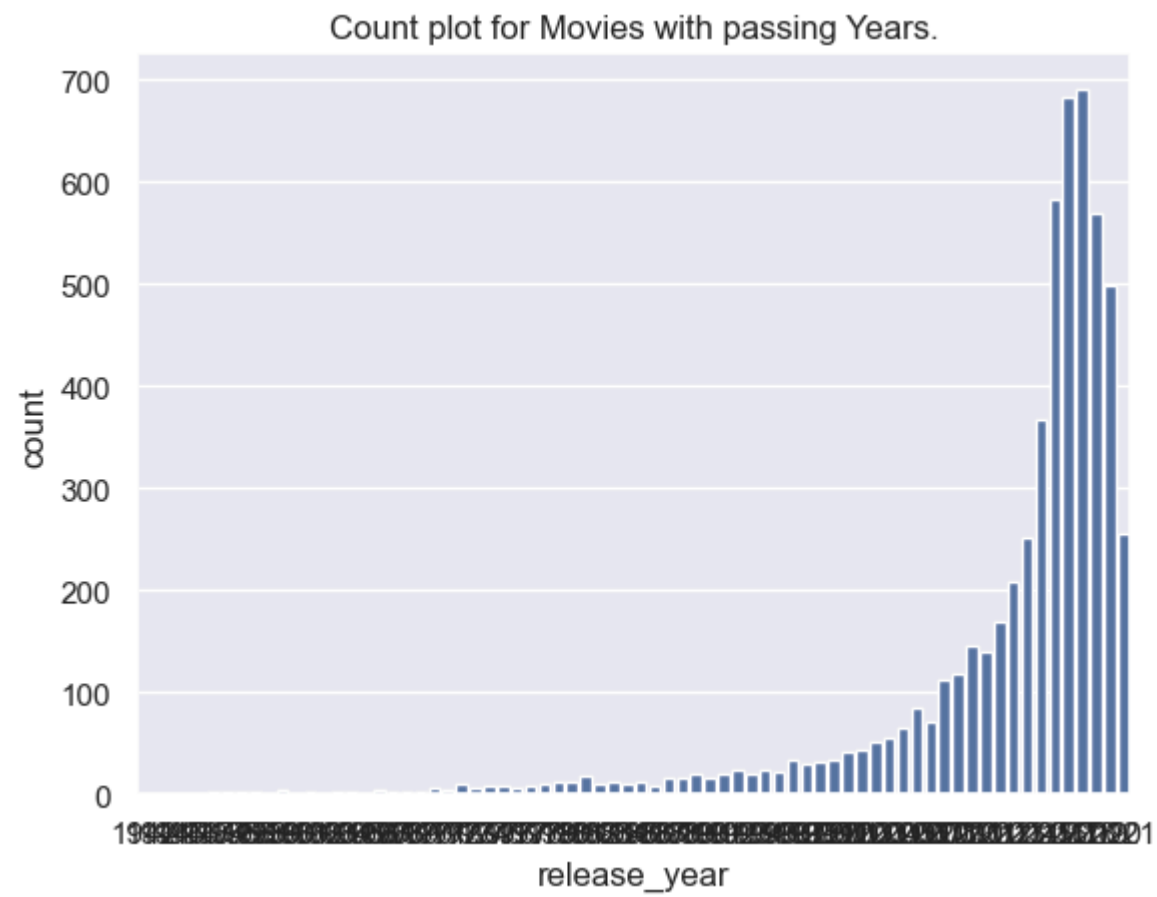
```
Out[43]: release_year
2018      691
2017      684
2016      584
2019      570
2020      499
...
1946       1
1963       1
1961       1
1959       1
1966       1
Name: release_year, Length: 72, dtype: int64
```

Observation:

From the above we can see that with passing years more movies are being made year by year

The above data shows that there is a sudden increase in the creation of movies in year 2016 as compared to year 2015

```
In [44]: sns.countplot(x='release_year', data=netflix).set_title('Count plot for Movies with passing Years.')
sns.set(rc={'figure.figsize':(60,20)})
plt.show()
```

```
In [45]: netflix.head()
```

Out[45]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description	Genre
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	To protect his family from a powerful drug lor...	Crime Thriller
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	The arrival of a charismatic young priest brin...	Drama
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Equestria's divided. But a bright-eyed hero be...	Children Fantasy Musical
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	On a photo shoot in Ghana, an American model s...	Drama
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	A talented batch of amateur bakers face off in...	Baking Reality Thriller

In [46]: `netflix.head()`

Out[46]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description	Genre
2	2	s3 TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	To protect his family from a powerful drug lor...	Crime Thriller
5	5	s6 TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	The arrival of a charismatic young priest brin...	Drama
6	6	s7 Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	missing	2021-09-24	2021	PG	91 min	Equestria's divided. But a bright-eyed hero be...	Children Fantasy Musical
7	7	s8 Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	On a photo shoot in Ghana, an American model s...	Drama
8	8	s9 TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	A talented batch of amateur bakers face off in...	Baking Reality Thriller

In [47]: `netflix['duration'].unique()`

```
Out[47]: array(['1 Season', '91 min', '125 min', '9 Seasons', '104 min', '127 min',
'94 min', '161 min', '61 min', '166 min', '147 min', '103 min',
'97 min', '106 min', '111 min', '110 min', '105 min', '96 min',
'124 min', '116 min', '98 min', '115 min', '122 min', '99 min',
'88 min', '100 min', '102 min', '93 min', '95 min', '85 min',
'83 min', '113 min', '13 min', '182 min', '48 min', '145 min',
'87 min', '90 min', '117 min', '128 min', '119 min', '143 min',
'114 min', '118 min', '108 min', '63 min', '121 min', '142 min',
'154 min', '120 min', '82 min', '109 min', '101 min', '86 min',
'229 min', '76 min', '89 min', '156 min', '112 min', '107 min',
'129 min', '92 min', '135 min', '136 min', '165 min', '150 min',
'133 min', '84 min', '140 min', '2 Seasons', '64 min', '59 min',
'139 min', '148 min', '189 min', '141 min', '130 min', '138 min',
'7 Seasons', '132 min', '123 min', '65 min', '68 min', '67 min',
'66 min', '62 min', '69 min', '70 min', '74 min', '131 min',
'46 min', '126 min', '155 min', '4 Seasons', '159 min', '137 min',
'12 min', '77 min', '49 min', '72 min', '78 min', '204 min',
'212 min', '25 min', '47 min', '81 min', '71 min', '149 min',
'15 min', '54 min', '80 min', '224 min', '162 min', '60 min',
'75 min', '79 min', '32 min', '158 min', '164 min', '173 min',
'181 min', '185 min', '73 min', '6 Seasons', '24 min', '51 min',
'151 min', '42 min', '22 min', '134 min', '177 min', '58 min',
'52 min', '53 min', '8 min', '57 min', '28 min', '50 min',
'26 min', '45 min', '171 min', '27 min', '33 min', '44 min',
'29 min', '146 min', '157 min', '203 min', '21 min', '30 min',
'194 min', '55 min', '5 Seasons', '15 Seasons', '233 min',
'237 min', '230 min', '195 min', '253 min', '152 min', '190 min',
'160 min', '23 min', '208 min', '180 min', '144 min', '174 min',
'170 min', '192 min', '209 min', '187 min', '172 min', '186 min',
'11 min', '8 Seasons', '193 min', '176 min', '17 min', '56 min',
'169 min', '40 min', '36 min', '20 min', '168 min', '153 min',
'214 min', '31 min', '163 min', '14 min', nan, '179 min', '38 min',
'3 Seasons', '43 min', '200 min', '196 min', '167 min', '41 min',
'37 min', '35 min', '178 min', '228 min', '18 min', '205 min',
'201 min', '191 min'], dtype=object)
```

```
In [48]: ##netflix_season = netflix[netflix["duration"]=="1 Season"]
```

```
##SAHIL
```

```
netflix_seasons_data = netflix[netflix['duration'].astype(str).str.contains('Season')]
```

```
In [49]: netflix_seasons_data
```

Out[49]:

Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description	
2	2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	missing	2021-09-24	2021	TV-MA	1 Season	To protect his family from a powerful drug lord...
5	5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	missing	2021-09-24	2021	TV-MA	1 Season	The arrival of a charismatic young priest brings...
8	8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	2021-09-24	2021	TV-14	9 Seasons	A talented batch of amateur bakers face off in...
11	11	s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanarot, Sushar Manaying, Pavarit M...	missing	2021-09-23	2021	TV-MA	1 Season	Struggling to earn a living in Bangkok, a man ...
47	47	s48	TV Show	The Smart Money Woman	Bunmi Ajakaiye	Osas Ighodaro, Ini Dima-Okojie, Kemi Lala Akin...	missing	2021-09-16	2020	TV-MA	1 Season	Five glamorous millennials strive for success ...
...	...	...	...	...	...	...	...	...	...	...	...	...
8172	8172	s8173	TV Show	Th Eena Meena Deeka Chase Comedy Show	Dheeraj Berry	Sanjay Keni, Priya Raina	India	2018-03-31	2015	TV-Y7	1 Season	A hungry fox and three resourceful surprising...
8214	8214	s8215	TV Show	The Blue Planet: A	Alastair Fothergill	David Attenborough	United Kingdom	2015-10-10	2001	TV-G	1 Season	David Attenborough

	Unnamed: 0	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	description
				Natural History of the Oceans								narrates this definitive ex...
8293	8293	s8294	TV Show	The Fear	Michael Samuels	Peter Mullan, Anastasia Hille, Harry Lloyd, Pa...	United Kingdom	2017-03-31	2012	TV-MA	1 Season	Richie Becket is an aging Brighton crime boss..
8302	8302	s8303	TV Show	The Five	Mark Tonderai	Tom Cullen, O.T. Fagbenle, Lee Ingleby, Sarah ...	United Kingdom	2017-09-15	2016	TV-MA	1 Season	Twenty years after 5-year old Jesse disappears..
8599	8599	s8600	TV Show	Toast of London	Michael Cumming	Matt Berry, Robert Bathurst, Doon Mackichan, S...	United Kingdom	2017-09-01	2015	TV-MA	3 Seasons	After a divorce and fatal career move, a class...

178 rows × 15 columns

```
In [50]: netflix['duration'].head()
```

```
Out[50]: 2      1 Season
5      1 Season
6      91 min
7     125 min
8      9 Seasons
Name: duration, dtype: object
```

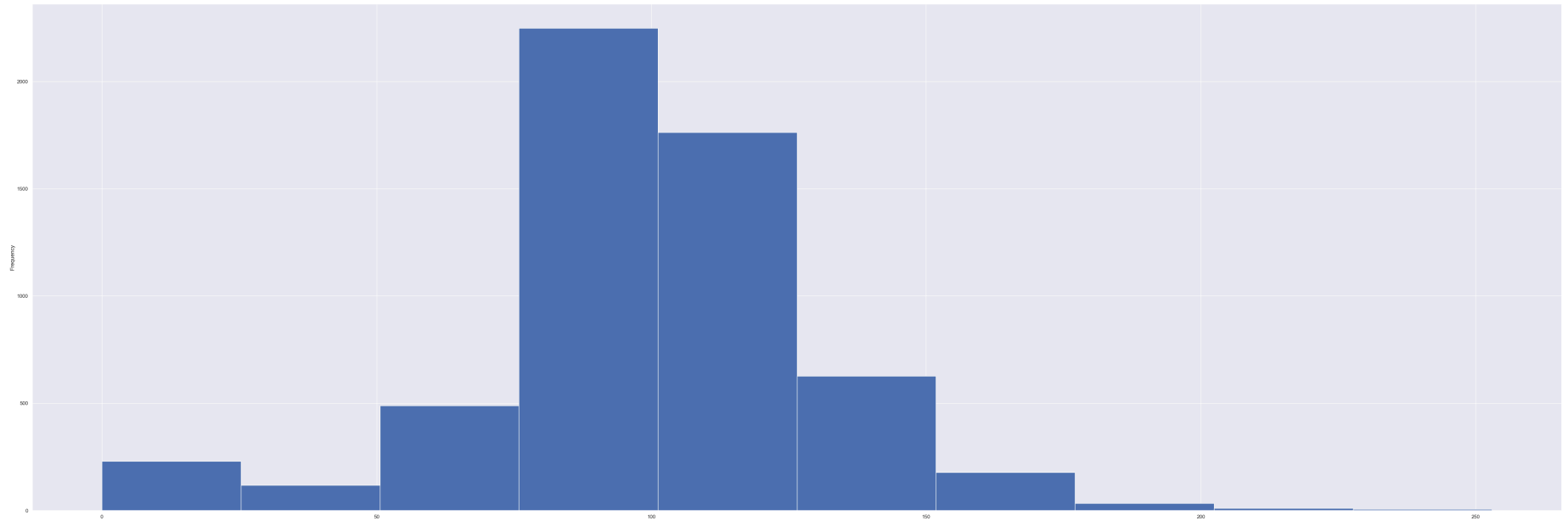
```
In [55]: netflix['duration'] = netflix['duration'].str.replace(' min', '')

netflix['duration'] = pd.to_numeric(netflix['duration'], errors='coerce')
```

```
netflix['duration'] = netflix['duration'].fillna(0).astype(int)
```

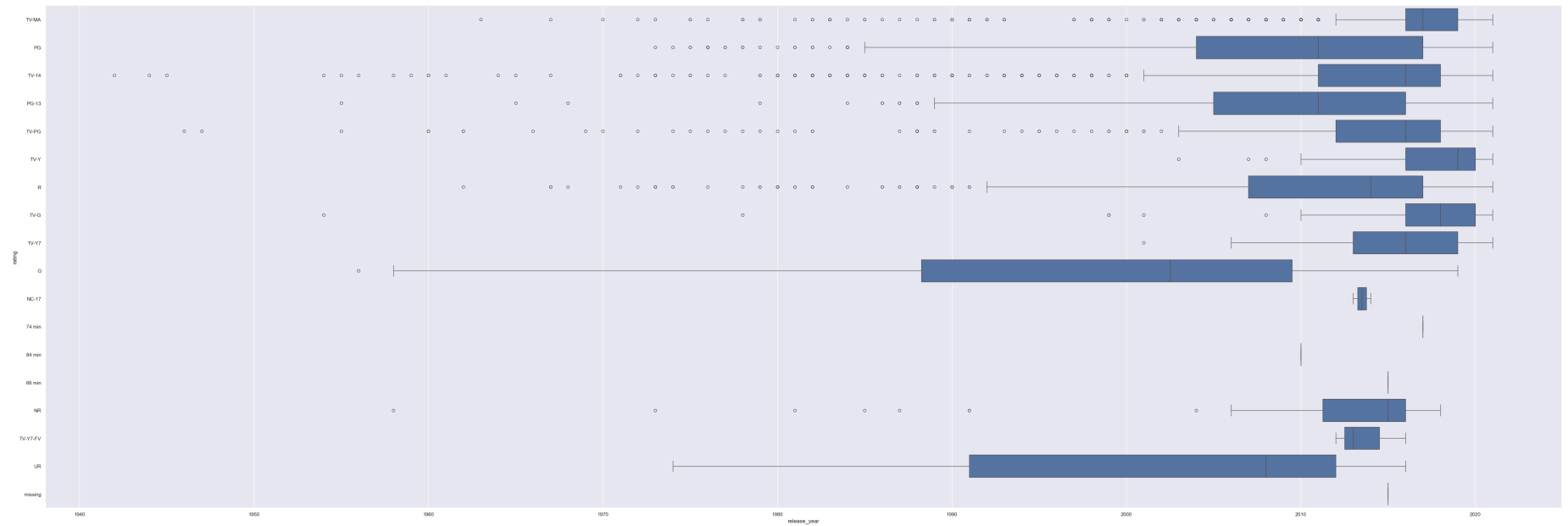
```
In [56]: netflix['duration'].sort_values().plot.hist()
```

```
Out[56]: <Axes: ylabel='Frequency'>
```



```
In [53]: sns.boxplot(x="release_year", y="rating", data=netflix)
```

```
Out[53]: <Axes: xlabel='release_year', ylabel='rating'>
```

```
In [54]: netflix[netflix['Genre 2'] == 'Action'].release_year.groupby(netflix.release_year).count().plot(kind='pie', figsize=(6, 6), aut
plt.axis('equal')
plt.show()
```

release_year

In []: